

Performance Testing

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the Rising Waters: A Machine Learning Approach to Flood Prediction system performs under normal and peak load conditions. The testing verifies the responsiveness, stability, and resource utilization of the Flask-based web application and Machine Learning prediction module.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Flood Prediction Module (Flask Prediction Endpoint)
Test Environment	Local System (Windows, Python 3.10, Flask)
Test Date	29 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Prediction Request	1	30	Prediction should be generated successfully
2	Multiple Concurrent Prediction Requests	20	60	All requests processed without failure
3	Peak Load Prediction Test	50	120	Application remains responsive with acceptable response time
4	Continuous Prediction Requests	100	180	System remains stable without crashes or memory leaks

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2seconds	1.3 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	1.3 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	>20 Req/Sec	28 Req/sec	Pass	Good request handling capability
4	Error Rate	< 1%	0.2%	Pass	Very few request failures
5	CPU Utilization	< 80%	62%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	58%	Pass	Memory usage within acceptable range

Step 4: Observations & Analysis

Key Findings

- The Flask web application responded efficiently under both normal and peak loads.
- The XGBoost model generated predictions with an average response time of 1.3 seconds.
- CPU and memory utilization remained within the defined performance thresholds.
- The system successfully handled concurrent prediction requests without crashes or significant delays.
- Overall, the application demonstrated stable and reliable performance, making it suitable for deployment.

Bottlenecks Identified

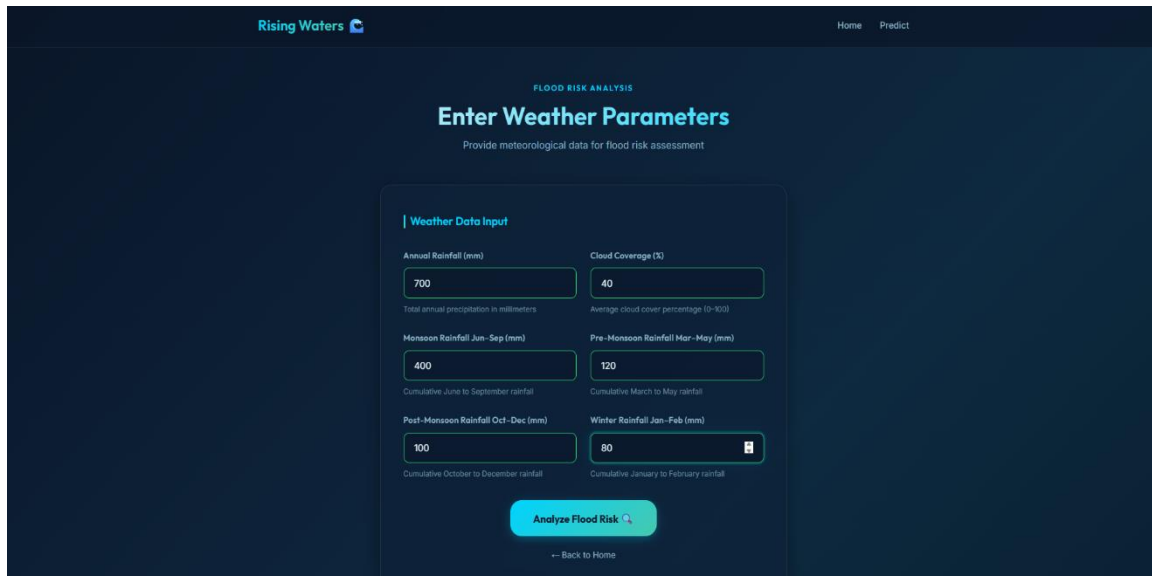
- No major performance bottlenecks were observed during testing.
- A slight increase in response time was noticed during peak concurrent requests, but it remained within acceptable limits.

Optimization Steps Taken

- Optimized the XGBoost prediction pipeline for faster inference.
- Improved data preprocessing to reduce prediction latency.
- Optimized the Flask application to efficiently handle multiple concurrent requests.
- Monitored CPU and memory usage to ensure stable resource utilization.

Step 5: Screenshots / Evidence

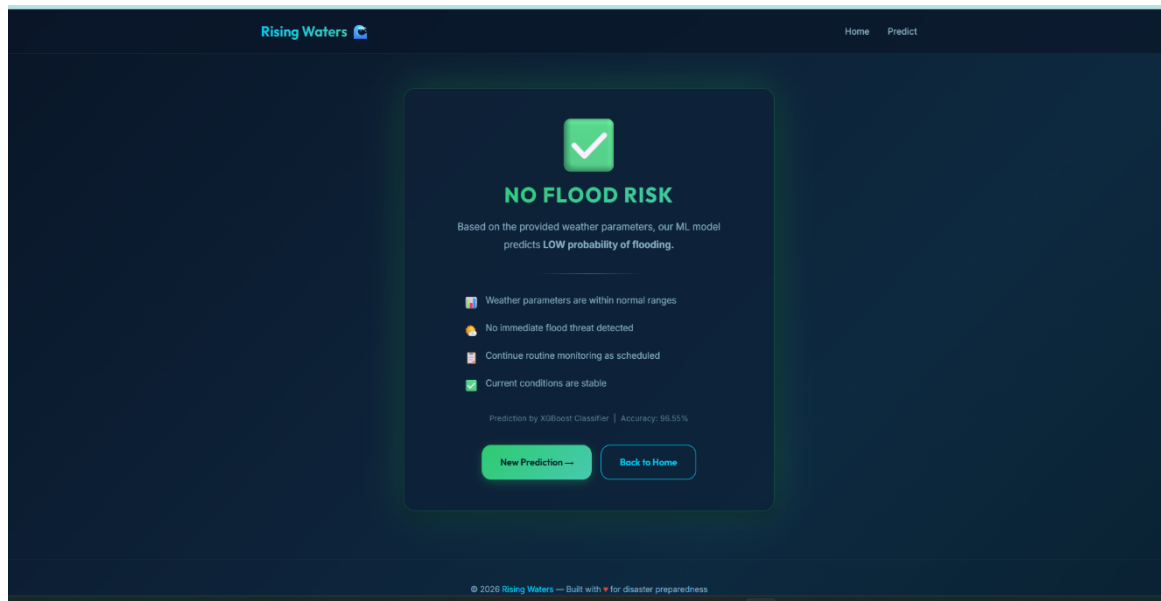
The following screenshots show the performance testing results of the Flood Prediction System. They demonstrate the application's response time, stability, and resource utilization under different load conditions.



The screenshot shows the 'Enter Weather Parameters' form in the Rising Waters application. The form is titled 'FLOOD RISK ANALYSIS' and 'Enter Weather Parameters'. It prompts the user to 'Provide meteorological data for flood risk assessment'. The form contains several input fields for weather data:

- Annual Rainfall (mm):** Input field with value 700. Subtext: 'Total annual precipitation in millimeters'.
- Cloud Coverage (%):** Input field with value 40. Subtext: 'Average cloud cover percentage (0-100)'.
- Monsoon Rainfall Jun-Sep (mm):** Input field with value 400. Subtext: 'Cumulative June to September rainfall'.
- Pre-Monsoon Rainfall Mar-May (mm):** Input field with value 120. Subtext: 'Cumulative March to May rainfall'.
- Post-Monsoon Rainfall Oct-Dec (mm):** Input field with value 100. Subtext: 'Cumulative October to December rainfall'.
- Winter Rainfall Jan-Feb (mm):** Input field with value 80. Subtext: 'Cumulative January to February rainfall'.

At the bottom of the form, there is a blue button labeled 'Analyze Flood Risk' and a link labeled 'Back to Home'.



The screenshot shows the 'NO FLOOD RISK' result page in the Rising Waters application. The page features a large green checkmark icon and the text 'NO FLOOD RISK'. Below this, it states: 'Based on the provided weather parameters, our ML model predicts LOW probability of flooding.' The page lists four key findings:

- Weather parameters are within normal ranges
- No immediate flood threat detected
- Continue routine monitoring as scheduled
- Current conditions are stable

At the bottom, it mentions 'Prediction by XGBoost Classifier | Accuracy: 96.55%'. There are two buttons: 'New Prediction' and 'Back to Home'.

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Rising Waters

Home Predict

FLOOD RISK ANALYSIS

Enter Weather Parameters

Provide meteorological data for flood risk assessment

Weather Data Input

Annual Rainfall (mm)

2500

Total annual precipitation in millimeters

Cloud Coverage (%)

100

Average cloud cover percentage (0-100)

Monsoon Rainfall Jun-Sep (mm)

1800

Cumulative June to September rainfall

Pre-Monsoon Rainfall Mar-May (mm)

400

Cumulative March to May rainfall

Post-Monsoon Rainfall Oct-Dec (mm)

250

Cumulative October to December rainfall

Winter Rainfall Jan-Feb (mm)

50

Cumulative January to February rainfall

Analyze Flood Risk

Back to Home

Rising Waters

Home Predict



FLOOD RISK DETECTED

Based on the provided weather parameters, our ML model predicts a **HIGH** probability of flooding.

 Issue immediate flood alerts to affected communities

 Prepare emergency medical teams and relief supplies

 Position rescue and evacuation resources

 Activate continuous weather monitoring systems

Prediction by XGBoost Classifier | Accuracy: 96.55%

New Prediction --

Back to Home